

Did Malta's PM2.5 death rate fall by two-thirds?

Testing the EEA's own estimate against its data, its method and Malta's station measurements

THE CLAIM UNDER REVIEW

“... is estimated to have been reduced by 67.7% between 2005 and 2023 (from 143.5 to 46.3, respectively), resulting in 172 (95% CI: 131-192) attributable deaths in 2023.”

European Environment Agency, Air pollution country fact sheets 2025: quick country facts (1 Dec 2025).

The rate: deaths attributable to PM2.5 above 5 µg/m³ per 100 000 people aged 30+, Malta.

VERDICT: LARGELY SUPPORTED

Accurate report of a modelled estimate; 2005 start not measured

Version 1.0 · 6 October 2026

Status: no right of reply needed

Prepared from public sources and EEA and Eurostat data.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

The European Environment Agency’s *quick country facts* (page modified 1 December 2025) say that in Malta the rate of natural deaths attributable to long-term exposure to PM2.5 above 5 µg/m³, per 100 000 people aged 30 or over, “**is estimated to have been reduced by 67.7% between 2005 and 2023 (from 143.5 to 46.3, respectively), resulting in 172 (95% CI: 131–192) attributable deaths in 2023.**”

- 1

The figures are the EEA’s own, reported exactly. The EEA’s table gives 143.5 (2005) and 46.3 (2023), a fall of 67.74%, and 172 deaths (95% CI 131–192) in 2023, as does Eurostat. The technical report rounds to 173 (132–193): one death.
- 2

The method is the documented one, and it reproduces. 8% more deaths per 10 µg/m³ (WHO), counted above 5 µg/m³, applied to natural deaths of people aged 30+. From Eurostat’s death counts we get 346 and 170 deaths for 2005 and 2023, within 1.2% of the EEA’s 348 and 172.
- 3

Measurements agree from 2007; 2005 has none. Malta’s PM2.5 records in the EEA archive begin in August 2006, so the 2005 value (21.0 µg/m³) is a map value. From 2007 the stations fall at about the map’s pace, and in 2023 five stations average 10.9 µg/m³, the map’s own figure.
- 4

It is a rate, and the start and threshold matter. The number of deaths fell by half (348 to 172) because the over-30 population grew 54%. Counting all concentrations, or starting in 2007, the fall is about 55–57%.
- 5

Verdict: largely supported (high confidence). The statement accurately reports the EEA’s estimate and says it is an estimate. The caveat: the exact 67.7% rests on a 2005 value that was not measured in Malta, and the 95% CI covers only the risk function.

THIS CLAIM

SUPPORTED

LARGELY SUPPORTED

NOT SUBSTANTIATED

MISLEADING

CONTRADICTED

67.7%

fall in the modelled rate, 2005–2023: reproduced from the EEA’s table

172

attributable deaths in 2023 (95% CI 131–192); the technical report says 173

–51%

fall in the number of deaths (348 → 172); the over-30 population grew 54%

0

Malta stations reporting PM2.5 to the EEA in 2005: that year’s value comes from a map

WHAT WOULD MOVE THE VERDICT UP

Evidence that Malta’s 2005 map value holds: an EEA uncertainty range for Malta’s exposure, or a check of the 2005 map against the PM10 that Malta did measure that year.

WHAT WOULD MOVE IT DOWN

An EEA revision of the series, or evidence that the 2005 map was off by much more than its typical error (about 3 µg/m³), so that the 2005–2023 fall changes materially.

IN THIS REPORT	
1 The claim and what we could verify	5 Where the evidence points different ways
2 Method	6 Testing the claim
3 How the estimate is made	7 Verdict and requests for evidence
4 What the measurements show	8 Limitations

1 The claim and what we could verify

The statement is the EEA’s own text on the quick-facts page of its 2025 air pollution country fact sheets, read in the page HTML on 6 October 2026 [1]. The page gives a sentence of the same form for each country it covers. The EEA is the EU’s environment agency; we assess the wording of its statement, not the agency. Because it is the EEA’s own statistic, the questions are whether the sentence reports the EEA’s estimate accurately, and what that estimate does and does not mean.

What	Where	Access
The Malta sentence (quoted on the cover and in the TL;DR)	Quick country facts [1]	Read in full, 6 Oct 2026
The estimates by year and scenario, with population and concentration	EEA burden-of-disease table [2]; Eurostat sdg_11_52 [3]	Downloaded, 6 Oct 2026
The method and its history	EEA briefing [4] and indicator [5]; ETC HE technical report [6]; map reports for 2005 [7] and 2023 [8]	Read in full
Measured PM2.5 and PM10 in Malta, 2004–2025	EEA station files [10]; PQ 29696 table via CC-007 [15]	Downloaded; reused

Our intake record first paraphrased the statement as “PM2.5 deaths down two-thirds”. The EEA speaks of the *rate*, and the number of deaths fell by about half (section 6), so the record title is now “PM2.5 death rate down two-thirds”. The EEA’s *Europe’s environment 2025* page for Malta draws the same series as a chart, with data to 2022 [14]; it is context, not the statement checked.

SCOPE NOTE

This check tests the EEA’s statement about its own estimate. It does not assess Malta’s air-quality policy or any authority, and it does not count deaths: attributable deaths are a statistical estimate for a population, not deaths certified as caused by air pollution (the EEA explains the difference in material linked from its briefing [4]).

2 Method

Question. Do 143.5, 46.3, 67.7% and 172 (131–192) match the EEA’s own estimates; is the method as described; is the 2005 starting point comparable with 2023; and do Malta’s measured concentrations bear out the modelled trend?

Evidence. We downloaded the EEA’s *Burden of disease of air pollution (Countries & NUTS)* table through its Table publisher [2]; Malta for every scenario, all countries, and the EU-27. We downloaded Eurostat sdg_11_52 (the same estimates [3]), Eurostat population, deaths and causes of death [11] and emissions [12], and every Malta PM2.5 and PM10 file in the EEA’s station archive [10], all on 6 October 2026 with Eurostat’s flags (none on the Malta values used). *tools/cc-035-report/calc.py* recomputes every figure from *data/cc-035/*. We read the EEA’s briefing and indicator [4, 5], the technical report behind the estimates [6] and the two map reports [7, 8], and searched Crossref and Europe PMC for peer-reviewed work on the risk function and on Malta’s PM2.5 [9, 13].

Grades. Official estimates and statistics are grade C; the meta-analysis behind the risk function and Malta’s source study are grade B. **Verdicts** follow the five-point scale in Appendix A.

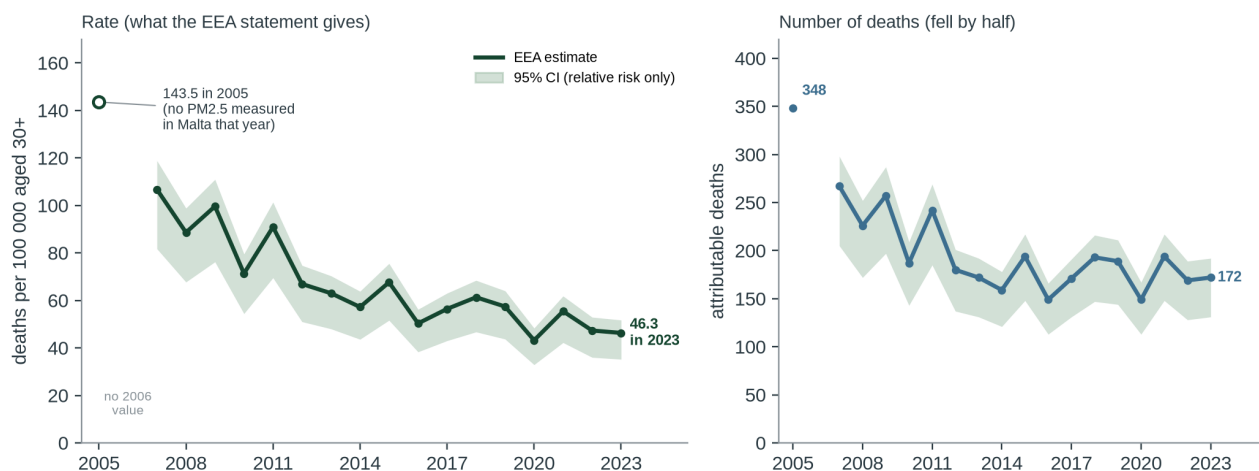
3 How the estimate is made

The EEA's estimate combines four things, cell by cell on a 1 km grid, and adds up the cells [6]:

- **Concentration.** An annual PM2.5 map that merges station measurements with a chemical transport model and other data (altitude, meteorology, population) [6, 8]. For Malta in 2023 its population-weighted mean is 10.9 $\mu\text{g}/\text{m}^3$ [6].
- **Population.** Gridded population scaled to Eurostat totals; for all-cause deaths, people aged 30 or over (373,208 in Malta in 2023; Eurostat 373,224 [2, 11]).
- **Baseline deaths.** Deaths from natural causes (all causes except injuries and other external causes) by age and sex, from Eurostat. Cause-of-death data begin in 2011, so 2011 stands in for 2005–2010 [6].
- **Risk.** 8% more natural deaths per 10 $\mu\text{g}/\text{m}^3$ (95% CI 6–9%), the WHO 2021 recommendation from Chen and Hoek's meta-analysis of 107 studies [6, 9], counted only above 5 $\mu\text{g}/\text{m}^3$, the WHO 2021 guideline level [4].

The attributable fraction in a cell is $1 - \exp(-\beta \times (C - 5))$, with $\beta = \ln(1.08)/10$, and the deaths are that fraction of the cell's baseline deaths [6]. Using Malta's national mean concentration and Eurostat's natural deaths aged 30+ [11], we get 346 deaths for 2005 and 170 for 2023, against the EEA's 348 and 172 (–0.6% and –1.2%). The small gap is expected, since the EEA works cell by cell [4].

What changed in the method. Until 2021 the EEA used a risk of 6.2% per 10 $\mu\text{g}/\text{m}^3$ counted from 0 $\mu\text{g}/\text{m}^3$; from 2022 it uses 8% counted above 5 $\mu\text{g}/\text{m}^3$, and since 2024 it gives rates per person aged 30 or over. It recalculated every year back to 2005 with the current method [5]. The 95% CI reflects the uncertainty in the risk only; “other possible uncertainties (for instance, the one related to exposure assessments) are not quantified” [4].



Deaths from all natural causes, people aged 30+, attributed to PM2.5 above 5 $\mu\text{g}/\text{m}^3$ (WHO 2021 guideline), EEA burden-of-disease table, retrieved 6 Oct 2026. The rate fell 67.7% while the population aged 30+ grew 54% (242,534 to 373,208), so the number fell 50.6%. The EEA gives no 2006 value.

Figure 1. The EEA's estimate for Malta. Left: the rate the statement gives, with its 95% CI. Right: the number of attributable deaths. The rate fell 67.7%; the number fell 50.6% because more people were aged 30 or over. The EEA gives no value for 2006; 2005 is drawn hollow because no PM2.5 was measured in Malta that year [2, 10].

4 What the measurements show

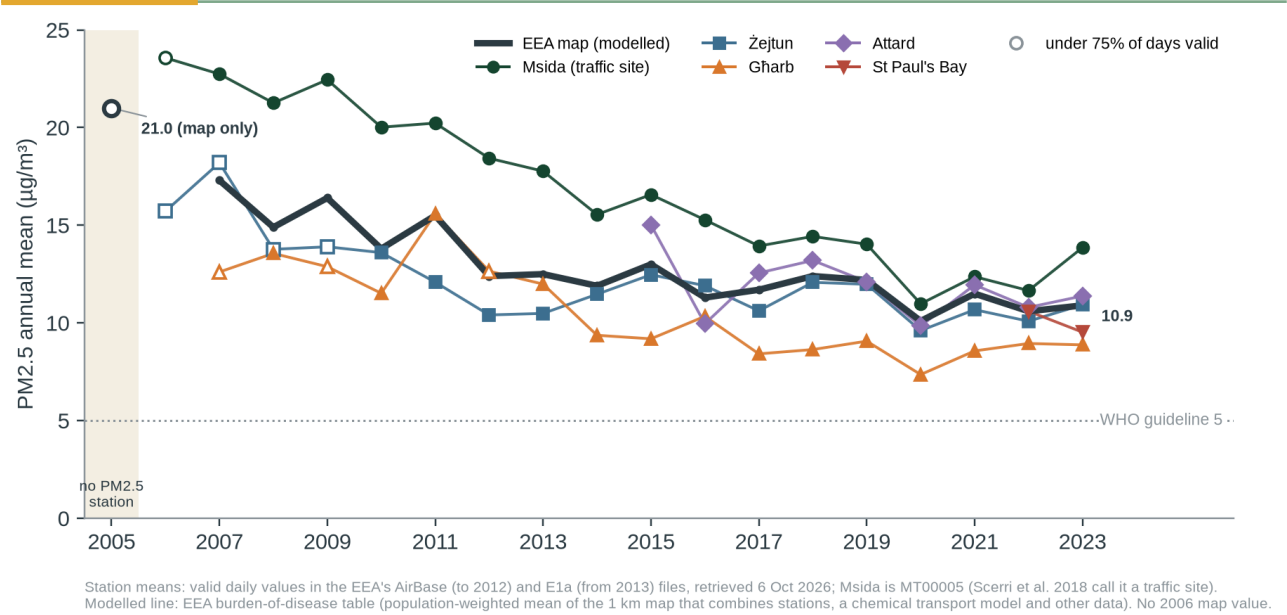


Figure 2. The EEA’s modelled population-weighted PM2.5 for Malta (thick line) and the annual means at Malta’s stations (thin lines; hollow markers where fewer than 75% of days have valid data) [2, 10].

2005. The EEA’s station archive has no PM2.5 measurement in Malta before 29 August 2006 (Żejtun), then Msida from October 2006 and Għarb from June 2007 [10]. In 2005 Malta measured PM10 at two stations (annual means 49.0 and 42.4 µg/m³ at MT00002 and Kordin) [10]. The 2005 reference map that the EEA uses was built from 173 PM2.5 stations across Europe, about 1,410 “pseudo” PM2.5 values estimated from PM10, and the EMEP model; it gives Malta 21.2 µg/m³ [7] (21.0 in the current table [2]). Its typical error in cross-validation was 2.9 µg/m³ in urban background areas [7].

From 2007. The measurements fall at about the map’s pace. At Msida, 22.7 µg/m³ in 2007 became 13.9 in 2023 (–39%); the map went from 17.3 to 10.9 (–37%). In the 13 years when Msida, Żejtun and Għarb all have at least 75% of days (2010, 2011, 2013–2023), their mean sits between 1.0 below and 1.3 above the map, and fell 25% from 2010 to 2023 against the map’s 21% [2, 10]. In 2023 the five stations in the parliamentary table used in CC-007 average 10.9 µg/m³, the map’s value [6, 10, 15].

Context. Malta’s reported primary PM2.5 emissions fell from 750 to 310 tonnes between 2005 and 2023, with public electricity and heat production down from 420 to 3 tonnes [12]. Emissions are not the whole story: at Msida in 2016, Saharan dust and sea salt made up about a third of PM2.5 [13], and the risk function treats all PM2.5 mass alike [9].

Indicator	Malta	Comparison	Source	Grade
Rate per 100 000 aged 30+, 2005 → 2023	143.5 → 46.3	EU-27 153.0 → 59.7	[2]	C
Fall in the rate, 2005–2023	67.7%	EU-27 61.0%; Malta 14th of 27	calculated	C
Attributable deaths, 2005 → 2023	348 → 172	EU-27 423,593 → 182,399	[2, 3]	C
Modelled population-weighted PM2.5 (µg/m³)	21.0 → 10.9	EU-27 19.4 → 10.2	[2]	C
Measured PM2.5, five stations, 2023 (µg/m³)	10.9	map 10.9	[10, 15]	C
Natural deaths aged 30+, baseline rate per 100 000	1,238 → 1,038	implied by the EEA figures; Eurostat 1,231 → 1,026	calculated [11]	C

All values in data/cc-035/checks.csv, station_vs_model.csv and eu27_rate_change.csv.

5

Where the evidence points different ways

Q1 Is the 2005 starting point comparable with 2023?

COMPARABLE BY DESIGN;
2005 NOT MEASURED

EVIDENCE FOR THE CLAIM'S IMPRESSION

The EEA recalculated every year back to 2005 with the current health method [5]; the 2005 map was made with the same mapping method as later maps [7]; the series uses one method for 2005–2023 [6]. Our approximation matches both ends within 1.2%. From 2007, Malta's stations fall at about the map's pace (section 4).

EVIDENCE AGAINST

No PM2.5 was measured in Malta in 2005 [10]. The 2005 map rests on PM10-based estimates and the EMEP model; the 2023 map on measured PM2.5 and the CAMS model ensemble [7, 8]. Shifting Malta's 2005 value by the map's typical error (2.9 µg/m³) gives a fall of 61% to 72% (a Europe-wide error applied to one country: indicative only). The 95% CI leaves exposure error out [4].

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: the 2005 value is plausible but unmeasured, so the fall is less certain than its one decimal place suggests. The direction, and most of the size, are borne out by measurements from 2007.

Q2 Does a 67.7% fall in the rate mean two-thirds fewer deaths?

ACCURATE AS WORDED

EVIDENCE FOR THE CLAIM'S IMPRESSION

The EEA says “rate”, names the threshold (above 5 µg/m³) and the age group, and says “is estimated”. A rate per person at risk is how the EEA compares countries [4]. Eurostat's rate per inhabitant of all ages fell 64% [3].

EVIDENCE AGAINST

The number of deaths fell 50.6% (348 to 172), because the over-30 population grew 54% [2]. Part of the rate's fall is a 16% lower baseline death rate among people over 30, not cleaner air (section 6). Counting all concentrations the fall is 54.7% [2]. And 5 µg/m³ is not a no-effect level: studies at low concentrations find similar or higher risks [9], and the EEA notes no evidence of a threshold [4].

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: the sentence answers one well-defined question correctly. It does not say that two-thirds fewer people died from PM2.5, and readers should not take it that way.

6

Testing the claim

Sub-claim	What the evidence shows	Rating
A. The rate was 143.5 per 100 000 aged 30+ in 2005	EEA table: 143.5 (348 deaths, 242,534 people aged 30+) [2]; Eurostat: 348 deaths [3].	ACCURATE
B. It was 46.3 in 2023	EEA table and technical report: 46.3 (35.2–51.7) [2, 6].	ACCURATE
C. “reduced by 67.7% between 2005 and 2023”	$(143.5 - 46.3) / 143.5 = 67.74\%$. The 2005 value is a map value with no PM2.5 measured in Malta [7, 10]; from 2007 the fall is 56.6%; the 95% CI covers the risk only [4].	LARGELY ACCURATE
D. “172 (95% CI: 131-192) attributable deaths in 2023”	EEA table and Eurostat: 172 (131–192) [2, 3]. Technical report: 173 (132–193) [6]; $46.3 \times 373,208 / 100\,000 = 172.8$: consistent with 172 truncated and 173 rounded.	ACCURATE
E. Natural deaths, long-term exposure above 5 µg/m³, people aged 30+	Matches the documented method [4–6]; recomputed from Eurostat deaths within 1.2% (section 3).	ACCURATE

Why the rate fell. Splitting the change: the attributable fraction fell from 11.6% to 4.4% of natural deaths ($\times 0.383$), as the modelled excess above 5 µg/m³ fell from 16.0 to 5.9 µg/m³; and the baseline natural death rate among people aged 30+ fell 16% ($\times 0.838$) [2, 11]. Together: $\times 0.321$, a fall of 68%. On the attributable fraction alone the fall would be 61.7%.

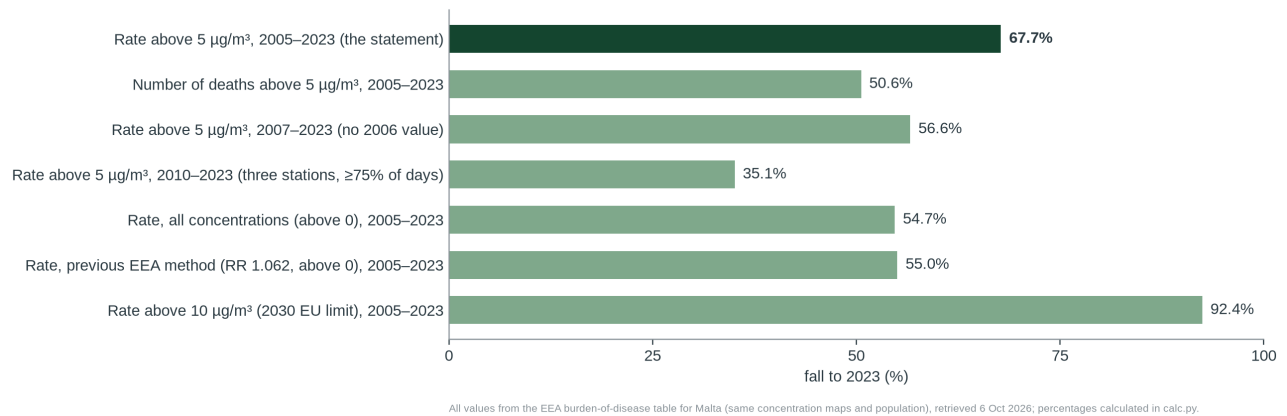


Figure 3. The same EEA estimates measured different ways. Only the top bar is the statement; the others show how the size of the fall depends on the threshold, the method, the start year and whether rates or numbers are compared [2].

7 Verdict and requests for evidence

VERDICT LARGELY SUPPORTED

The statement reports the EEA’s own estimate exactly and calls it an estimate; its 2005 starting value comes from a map, with no PM2.5 measured in Malta that year. Confidence: high.

Why. (1) Every figure in the sentence reproduces from the EEA’s own table and from Eurostat. (2) The method is as described, and we reproduce the deaths within 1.2% from Eurostat’s death counts. (3) The sentence is carefully worded: a rate, above 5 µg/m³, people aged 30+, “is estimated”. (4) Malta’s station measurements, an independent line, follow the modelled concentrations from 2007 and match them in 2023. (5) The caveat is the start: the 2005 value was not measured in Malta, and the 95% CI leaves exposure error out, so the exact 67.7% is less certain than it looks. That is a minor caveat; it does not change the substance, a large fall. Confidence is high because the EEA’s data, Eurostat, our recomputation and the measurements agree.

What this verdict does not say. It does not say that two-thirds fewer people died because of PM2.5 (the number fell by about half), that Malta’s air meets the WHO guideline (10.9 µg/m³ against 5 in 2023), or anything about the causes of the fall or any authority’s policy.

Evidence we are asking for

- ☐ From the EEA: an uncertainty range for Malta’s modelled exposure, above all for 2005, which the 95% CI does not include.
- ☐ From the EEA: confirmation that 172 (131–192) on the quick-facts page and 173 (132–193) in the technical report differ only by rounding, and why there is no value for 2006.
- ☐ From the Environment and Resources Authority, which runs Malta’s stations: any PM2.5 measurements from 2005 that were not reported to the EEA.

8 Limitations

- The statement is a modelled estimate; we tested it against the EEA’s own data, its documented method and measured concentrations, not against an independent mortality study for Malta (we found none).
- Our recomputation uses Malta’s national mean concentration, not the EEA’s 1 km grid, so it is a check on the order of the result, not a replication.

- The map errors we quote are Europe-wide cross-validation statistics, not Malta-specific; the 61–72% range is indicative.
- Station annual means are averages of valid days from the EEA's files, not the authority's official annual statistics; for 2023 they match the PQ 29696 table to 0.1 µg/m³. Station types were not taken from the EEA's metadata; only Msida is described (as a traffic site) by the source study [13].
- Chen and Hoek [9] and Scerri et al. [13] were read as abstracts.
- The quick-facts and Malta pages were not archived (the Wayback Machine refused requests on 6 October 2026).

References

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- [15] Mižien. Claim Check 007: Within EU limits vs WHO guideline (PQ 29696 station table). data/cc-007/.
- [16] Mižien. Data and calculations: data/cc-035/; tools/cc-035-report/calc.py.

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand (none in this report).

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	6 Oct 2026	First issue. Verdict Largely supported (high confidence); no right of reply needed.